

Week	Goal	Main Tasks
1	Learn Classical HMM fundamentals and theory	Learn about Markov chains, transition probabilities, emission distributions, and initial-state probabilities. Understand how posterior state probabilities are computed.
2	Learn the Training Algorithm for HMM	Understand Forward and Backward algorithms, Viterbi decoding, and Baum–Welch/Expectation-Maximization training.
3	Implement a classical Gaussian HMM & Prepare financial data and experimental setup	Implement or reproduce a simple HMM. Apply a Gaussian-emission HMM to basic sample data and inspect decoded states and posterior probabilities. Collect and preprocess S&P 500 daily data. Compute suitable market features.
4	Establish Gaussian HMM baseline & Review Gaussian Mixture Models	Train and evaluate the Gaussian HMM baseline. Review Gaussian Mixture Models and their EM-based training procedure.
5	Compare Gaussian and GMM-HMM models	Train and evaluate GMM-HMM. Run controlled experiments using identical data splits and state configurations and compare GMM emissions with the Gaussian baseline.
6	Midterm Report	Write up report and create midterm project demo.
7	Learn Fundamentals for Normalizing Flows	Go over change-of-variables theorem, invertible transformations, Jacobian determinants, maximum-likelihood training, and base distributions. Learn the theory behind Non-linear Independent Components Estimation (NICE).
8	Understand RealNVP and Implement RealNVP density estimation	Learn about the extension of NICE: Real-valued Non-Volume Preserving (RealNVP) flows. Implement and test RealNVP density estimations.
9	Integrate RealNVP with HMM Inference	Construct state-conditioned RealNVP emission models. Integrate RealNVP emission likelihoods with forward-backward inference. Implement a gradient-based or generalized-EM training procedure and compare initial results with Gaussian and GMM-HMM baselines.
10	Final Evaluations Across Emission Models	Perform controlled Gaussian/GMM/RealNVP-HMM experiments using identical data splits, state counts, and random seeds. Evaluate held-out likelihood, regime characteristics, persistence, transition behavior, stability, and correspondence with historical market events. Conduct a simple downstream risk/trading evaluation if time permits.
11	Final Report and Demo	Work on the final report and project demo.
12	Final Report and Demo	Finish the final report and complete the final project demo.